Reading a Vernier Caliper

PHY 118 · FLCC Physics · Dr. Byungmoon Cho (adapted from the original by Prof. T. Johnson-Steigelman)

What the two scales do

A Vernier scale lets you read one extra digit of precision without ever guessing. The caliper carries two scales: a fixed main scale and a shorter moving scale whose divisions are slightly smaller than the fixed scale's. Our classroom calipers read to 1/50 of a millimeter — that is, ±0.002 cm. Read the metric scale (cm), not the inch scale, and confirm the units before you record anything.



Figure 1. A reading on the classroom caliper. A marks the zero of the moving scale (the pointer); B marks the one moving-scale line that lines up with a fixed-scale mark.

Reading in two steps

Step 1 — the fixed scale. The zero mark of the moving (Vernier) scale is your pointer. Read the last mark it has passed on the fixed cm scale. In Figure 1 the moving-scale zero (A) sits just past 4.1 cm, so the fixed-scale part of the reading is 4.1 cm.

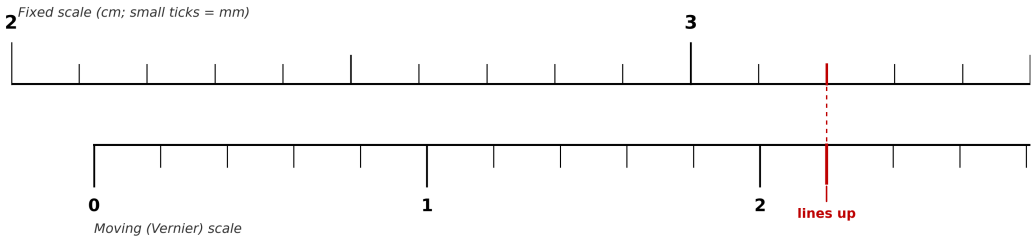
Step 2 — the last digits from the Vernier scale. Find the one place where a line on the moving scale lines up exactly with a line on the fixed scale — on a good caliper it happens at only one spot. Count the moving scale's own zero as the start, then read that aligned line. In Figure 1 the "6.4" line (B) is the one that coincides, which adds 0.064 cm, so the full reading is 4.1 cm + 0.064 cm = 4.164 cm. Trust only the line that truly coincides; how far the moving-scale zero sits past the last fixed mark tells you roughly how large that fraction should be, which is a good check.

Make it make sense

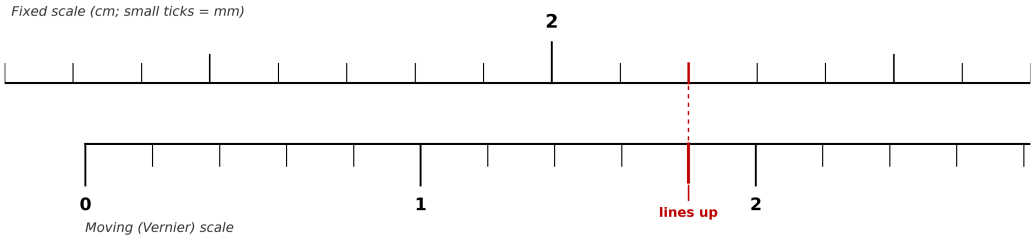
When you are new to the caliper, check a reading or two against a ruler. It is easy to be off by a whole millimeter or centimeter, or to read the inch scale by mistake, so confirm the units and ask whether the size is reasonable before you write it down. If you cannot spot the aligned line, look farther along the scale — it is usually obvious once you find it. A caliper reading takes a little longer than a ruler reading, but you gain a precise digit you could never estimate, and that trade is the whole point.

Try it out

Read the measurement on each Vernier scale below.



→ 2.122 cm



→ 1.318 cm

Adapted from "Reading a Vernier Scale," © 2017 Trevor Johnson-Steigelman.